

BRAC UNIVERSITY
Department of Computer Science and Engineering

Quiz 04
Semester: Fall 2024

Time: 25 min
Full Marks: 20

Name:	ID:	Section:
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CSE 251: Electronic Devices and Circuits

1. [CO1] 5

a) **What** is the S model of a MOSFET. **Explain** briefly with I-V characteristics graph.

2. [CO3] 5

Design a circuit using MOSFETs (s model) to implement the following logic function
$$f = (A+B).C$$

Analyze the circuit to find I_{DS} . For the Q_1 MOSFET $V_T = 1\text{V}$ and $K_n = 2\text{ mA/V}^2$. You must validate your assumption.

